Qus. Show the following formula logical eq^{nt} using eq^{nce} laws.

$$(P \land Q) \lor (P \land \sim Q) = P$$

Stochastic Annealing or Simulated Annealing:

⇒ World needs optimization..

→ Finding the best solution to complex problem can be challenging due to large search space and your search space is filled with local optima.

→ inspired by the physical process of annealing in metallurgy.

→ It is a probabilistic technique used for solving combinatorial or optimization problems.

Designed to search optimal or near optimal solution.
where material is heated and slowly cooled to remove defects & achieve a stable crystalline structure.

→ heat refers to degree of randomness in search space.
→ degree of randomness decreases over time to refine the solⁿ.
→ This technique can escape local maxima by introducing controlled randomness in search space

Steps:—

① Initialization:— Initial solⁿ = S_0
Initial temp = T_0

Here the temp. controls how you will accept the worst solution.

② Neighbourhood search: at each step s' new solution s' is generated, by making small changes in s.

③ Objective function Evaluation:—
s' is evaluated using objective fⁿ.
Now, if s' provides you the better solution then it is accepted.

④ Acceptance Probability:— If s' is worse than s, it may still be accepted with probability based on temperature and difference in objective function value.

difference in obj. fn.

$$P(\text{accept}) = e^{-\Delta E/T}$$

(5) cooling schedule:— After each iteration temp is decreased.

→ a according to a predefined cooling schedule.
→ It determines how quickly the algo converses.

(6) Termination: until reaches a low temp. algo continues or a predetermined no. of iteration is reached.

Adventages:—
→ ability to escape local minima.
→ Global optimization.
→ Flexibility — it can be applied to continuous and discrete optimization problem.

Limitation:—
- Parameter sensitivity
- computational time
- slow convergence.

Application:—
→ VLSI design
→ ML problem
→ scheduling problem
→ TSP
→ Protien folding app.

Genetic Algorithm:— Goldberg

↳ inspired by the darwin's theory of evolution

→ Solution generated by genetic algo is chromosomes (made of genes).
→ collection of chromosomes is called population.

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Steps:—

① Determine the no. of chromosomes, generation, mutation rate and cross over rates.

② Generate chromosome - chromosome number of the population and the initialization value of the genes.

③ Repeat 4-7 untill no of generation is met.

④ Evaluation of fitness value of chromosomes by calculating objective function.

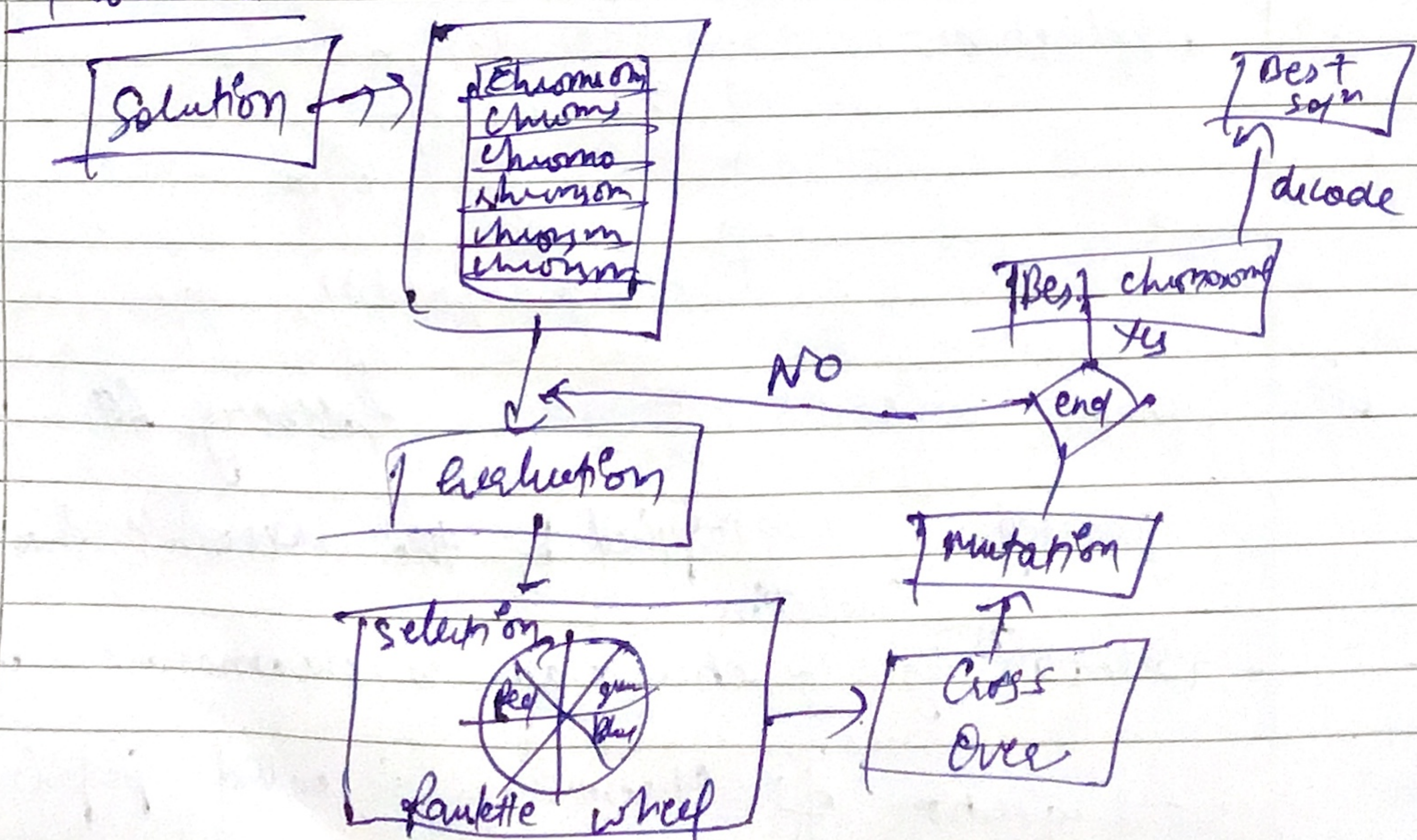
⑤ chromosome Selection.

⑥ Cross Over.

⑦ Mutation.

⑧ Solution (Best chromosome)

Flowchart:—



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Ques $a+2b+3c+4d=30$
genetic algo will be used to find a, b, c, d to satisfy the eqn:

$$F(x) = (a+2b+3c+4d) - 30.$$

| a | b | c | d | 0 - 30

(1) Ch [1] = [12, 5, 23, 8]
[2] = [2, 21, 18, 3]
[3] = [10, 4, 13, 14]
[4] = [20, 1, 10, 6]
[5] [1, 4, 13, 19]
[6] [20, 5, 17, 01]

(2) Evaluation:

F_obj [1] = 93
[2] = 80
[3] = 83
[4] = 46
[5] = 94
[6] = 55

(3) Fitness_score [1] = $\left(1/1+F\_obj[1]\right)$; $= \frac{1}{1+93} = 0.0106$
[2] = 0.0123
[3] = 0.0119
[4] = 0.0213
[5] = 0.0105
[6] = 0.0179

(4) Total = 0.0845

Probability of selection of each chromosome,
$P[1] = \frac{0.0106}{0.0845} = 0.1254$

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P [2] = 0.1456
[3] = 0.1408
[4] = 0.2521 ⟶ highest fitness
[5] = 0.1243
[6] = 0.2118

(5) Cumulative probability C [1] = 0.1254
C [2] = 0.1254 + 0.1456
C [3] = 0.1254 + 0.1456 + 0.1408
⋮
C [6]

(6) Random Numbers: R [1] = 0.201
R [2] = 0.284
[3] = 0.099
[4] = 0.822
[5] = 0.398
[6] = 0.501

↓

If random number R [1] is greater than C_1 and smaller than C_2 then select chromosome Ch[2] as chromosome in next gen.

new chromosome [1] = Ch [2]
[2] = Ch [3]
[3] = Ch [1]
[4] = Ch [6]
[5] = Ch [3]
[6] = Ch [4]

New population

(7) Cross Over